MATHS MANTHAN
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Subject: Mathematics **WEEKLY CHAPTER TEST – 05 (QUADRATIC EQUATIONS)** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. If $(a^2+b^2)x^2 - 2(ac+bd)x + (c^2+d^2) = 0$ has equal roots, then: [1]
(A) $ab = cd$ (B) $ad = bc$ (C) $ad = \sqrt{bc}$ (D) $ab = \sqrt{cd}$

Q2. The roots of the quadratic equation $x^2 + 3x - 10 = 0$ are: [1]
(A) $2, -5$ (B) $-2, 5$ (C) $2, 5$ (D) $-2, -5$

Q3. Which of the following quadratic equations has the sum of its roots as 4? [1]
(A) $2x^2 - 4x + 8 = 0$ (B) $-x^2 + 4x + 4 = 0$ (C) $\sqrt{2}x^2 - \frac{4}{\sqrt{2}}x + 1 = 0$ (D) $4x^2 - 4x + 4 = 0$

Q4. The quadratic equation $x^2 + x + 1 = 0$ has: [1]
(A) real and equal roots (B) irrational roots (C) real and distinct roots (D) no real roots

Q5. **Assertion (A):** If a and c are of opposite signs, then $ax^2 + bx + c = 0$ has real and distinct roots.
Reason (R): If a and c have opposite signs, $ac < 0 \implies -4ac > 0$, so $D = b^2 - 4ac > 0$. [1]
(A) Both A and R are true and R is correct explanation of A. (B) Both are true, but R is not correct explanation.
(C) A is true, R is false. (D) A is false, R is true.

SECTION B: Short Answer Type Questions (Part I)

Q6. If one root of the quadratic equation $2x^2 + kx - 6 = 0$ is 2, find the value of k and the other root. [2]

Q7. Solve the following equation for x: [3]

$$\frac{x-1}{x-2} + \frac{x-3}{x-4} = 3\frac{1}{3}, \quad x \neq 2, 4$$

SECTION B: Short Answer Type Questions (Continued) (Total Sec B = 11 Marks)

Q8. Solve for x by factorization: [3]

$$36x^2 - 12ax + (a^2 - b^2) = 0$$

Q9. A rectangular park is to be designed whose breadth is 3 m less than its length. Its area is to be 4 m^2 more than the area of a park in the shape of an isosceles triangle with base as the breadth of the rectangular park and altitude 12 m. Find the length and breadth of the rectangular park. [3]

SECTION C: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q10. Case Study: Agricultural Farm Parceling & Land Allocation

A progressive farmer owns a rectangular agricultural plot of total area 1260 m^2. He divides the land into two rectangular segments: a rice cultivation plot of dimensions x m × 30 m, and a wheat cultivation plot of dimensions $(x + 12)$ m × 30 m, such that the combined farm has a uniform width of 30 m and total length $(2x + 12)$ m.

Based on the above layout, answer the following questions:

(i) Formulate the algebraic equation representing the total area of the farm in terms of x. [1]

(ii) Solve the equation to find the value of x. [1]

(iii) Calculate the separate areas of the wheat field and the rice field. [2]

OR

Find the total cost of fencing the entire outer boundary of the 1260 m^2 farm at Rs. 40 per metre. [2]

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